

$$\begin{aligned}
C_{\text{ub}\phi_1} \frac{1}{16\pi^2} \rightarrow & \frac{1}{432} \mathbf{g}_1^3 y_u^{11i2} \text{LF}_{2,1,0} [\mathbf{m}_1, \mathbf{m}_q^{i1}] - \frac{1}{216} \mathbf{g}_1^3 y_u^{i1i2} \text{LF}_{3,1,-1} [\mathbf{m}_1, \mathbf{m}_{\bar{q}}^{i1}] + \\
& \frac{1}{432} \mathbf{g}_1^3 y_u^{i1i2} \text{LF}_{4,1,-2} [\mathbf{m}_1, \mathbf{m}_q^{i1}] + \frac{4}{27} \mathbf{g}_1^3 y_u^{i1i2} \text{LF}_{2,1,0} [\mathbf{m}_1, \mathbf{m}_{\bar{u}}^{i2}] - \\
& \frac{8}{27} \mathbf{g}_1^3 y_u^{i1i2} \text{LF}_{3,1,-1} [\mathbf{m}_1, \mathbf{m}_{\bar{u}}^{i2}] + \frac{4}{27} \mathbf{g}_1^3 y_u^{i1i2} \text{LF}_{4,1,-2} [\mathbf{m}_1, \mathbf{m}_{\bar{u}}^{i2}] + \\
& \frac{1}{16} \mathbf{g}_1 \mathbf{g}_2^2 y_u^{i1i2} \text{LF}_{2,1,0} [\mathbf{m}_2, \mathbf{m}_q^{i1}] - \frac{1}{8} \mathbf{g}_1 \mathbf{g}_2^2 y_u^{i1i2} \text{LF}_{3,1,-1} [\mathbf{m}_2, \mathbf{m}_q^{i1}] + \\
& \frac{1}{16} \mathbf{g}_1 \mathbf{g}_2^2 y_u^{i1i2} \text{LF}_{4,1,-2} [\mathbf{m}_2, \mathbf{m}_q^{i1}] + \frac{1}{9} \mathbf{g}_1 \mathbf{g}_3^2 y_u^{i1i2} \text{LF}_{2,1,0} [\mathbf{m}_3, \mathbf{m}_q^{i1}] - \\
& \frac{2}{9} \mathbf{g}_1 \mathbf{g}_3^2 y_u^{i1i2} \text{LF}_{3,1,-1} [\mathbf{m}_3, \mathbf{m}_q^{i1}] + \frac{1}{9} \mathbf{g}_1 \mathbf{g}_3^2 y_u^{i1i2} \text{LF}_{4,1,-2} [\mathbf{m}_3, \mathbf{m}_q^{i1}] + \\
& \frac{4}{9} \mathbf{g}_1 \mathbf{g}_3^2 y_u^{i1i2} \text{LF}_{2,1,0} [\mathbf{m}_3, \mathbf{m}_{\bar{u}}^{i2}] - \frac{8}{9} \mathbf{g}_1 \mathbf{g}_3^2 y_u^{i1i2} \text{LF}_{3,1,-1} [\mathbf{m}_3, \mathbf{m}_{\bar{u}}^{i2}] + \\
& \frac{4}{9} \mathbf{g}_1 \mathbf{g}_3^2 y_u^{i1i2} \text{LF}_{4,1,-2} [\mathbf{m}_3, \mathbf{m}_{\bar{u}}^{i2}] - \frac{1}{12} \mathbf{g}_1 \overline{y_d^{\text{pr}}} y_d^{i1r} y_u^{pi2} \text{LF}_{2,1,0} [\tilde{\mu}, \mathbf{m}_d^r] + \\
& \frac{1}{24} \mathbf{g}_1 \overline{y_d^{\text{pr}}} y_d^{i1r} y_u^{pi2} \text{LF}_{3,1,-1} [\tilde{\mu}, \mathbf{m}_d^r] + \frac{1}{24} \mathbf{g}_1 \overline{y_d^{\text{pr}}} y_d^{i1r} y_u^{pi2} \text{LF}_{4,1,-2} [\tilde{\mu}, \mathbf{m}_d^r] + \\
& \frac{1}{12} \mathbf{g}_1 \overline{y_u^{\text{pr}}} y_u^{pi2} y_u^{i1r} \text{LF}_{2,1,0} [\tilde{\mu}, \mathbf{m}_q^p] - \frac{5}{12} \mathbf{g}_1 \overline{y_u^{\text{pr}}} y_u^{pi2} y_u^{i1r} \text{LF}_{3,1,-1} [\tilde{\mu}, \mathbf{m}_q^p] + \\
& \frac{1}{3} \mathbf{g}_1 \overline{y_u^{\text{pr}}} y_u^{pi2} y_u^{i1r} \text{LF}_{4,1,-2} [\tilde{\mu}, \mathbf{m}_q^p] + \frac{1}{6} \mathbf{g}_1 \overline{y_u^{\text{pr}}} y_u^{pi2} y_u^{i1r} \text{LF}_{2,1,0} [\tilde{\mu}, \mathbf{m}_{\bar{u}}^r] - \\
& \frac{5}{24} \mathbf{g}_1 \overline{y_u^{\text{pr}}} y_u^{pi2} y_u^{i1r} \text{LF}_{3,1,-1} [\tilde{\mu}, \mathbf{m}_{\bar{u}}^r] + \frac{1}{24} \mathbf{g}_1 \overline{y_u^{\text{pr}}} y_u^{pi2} y_u^{i1r} \text{LF}_{4,1,-2} [\tilde{\mu}, \mathbf{m}_{\bar{u}}^r] - \\
& \frac{5}{108} \mathbf{m}_1 \mathbf{g}_1^3 a_u^{i1i2} \text{LF}_{2,1,1,0} [\mathbf{m}_1, \mathbf{m}_q^{i1}, \mathbf{m}_{\bar{u}}^{i2}] + \frac{1}{72} \mathbf{m}_1 \mathbf{g}_1^3 a_u^{i1i2} \text{LF}_{2,2,1,-1} [\mathbf{m}_1, \mathbf{m}_q^{i1}, \mathbf{m}_{\bar{u}}^{i2}] + \\
& \frac{5}{108} \mathbf{m}_1 \mathbf{g}_1^3 a_u^{i1i2} \text{LF}_{3,1,1,-1} [\mathbf{m}_1, \mathbf{m}_q^{i1}, \mathbf{m}_{\bar{u}}^{i2}] - \frac{1}{72} \mathbf{g}_1^3 y_u^{i1i2} \text{LF}_{1,1,1,0} [\mathbf{m}_1, \mathbf{m}_q^{i1}, \tilde{\mu}] + \\
& \frac{1}{36} \mathbf{g}_1^3 y_u^{i1i2} \text{LF}_{2,1,1,-1} [\mathbf{m}_1, \mathbf{m}_q^{i1}, \tilde{\mu}] - \frac{1}{72} \mathbf{g}_1^3 y_u^{i1i2} \text{LF}_{3,1,1,-2} [\mathbf{m}_1, \mathbf{m}_q^{i1}, \tilde{\mu}] - \\
& \frac{1}{72} \mathbf{m}_1 \mathbf{g}_1^3 a_u^{i1i2} \text{LF}_{2,2,1,-1} [\mathbf{m}_1, \mathbf{m}_{\bar{u}}^{i2}, \mathbf{m}_q^{i1}] + \frac{2}{9} \mathbf{g}_1^3 y_u^{i1i2} \text{LF}_{1,1,1,0} [\mathbf{m}_1, \mathbf{m}_{\bar{u}}^{i2}, \tilde{\mu}] - \\
& \frac{4}{9} \mathbf{g}_1^3 y_u^{i1i2} \text{LF}_{2,1,1,-1} [\mathbf{m}_1, \mathbf{m}_{\bar{u}}^{i2}, \tilde{\mu}] + \frac{2}{9} \mathbf{g}_1^3 y_u^{i1i2} \text{LF}_{3,1,1,-2} [\mathbf{m}_1, \mathbf{m}_{\bar{u}}^{i2}, \tilde{\mu}] - \\
& \frac{5}{144} \mathbf{g}_1^3 y_u^{i1i2} \text{LF}_{2,2,1,-2} [\mathbf{m}_1, \tilde{\mu}, \mathbf{m}_q^{i1}] + \frac{5}{36} \mathbf{g}_1^3 y_u^{i1i2} \text{LF}_{2,2,1,-2} [\mathbf{m}_1, \tilde{\mu}, \mathbf{m}_{\bar{u}}^{i2}] + \\
& \frac{1}{8} \mathbf{g}_1 \mathbf{g}_2^2 y_u^{i1i2} \text{LF}_{1,1,1,0} [\mathbf{m}_2, \mathbf{m}_q^{i1}, \tilde{\mu}] - \frac{1}{4} \mathbf{g}_1 \mathbf{g}_2^2 y_u^{i1i2} \text{LF}_{2,1,1,-1} [\mathbf{m}_2, \mathbf{m}_q^{i1}, \tilde{\mu}] + \\
& \frac{1}{8} \mathbf{g}_1 \mathbf{g}_2^2 y_u^{i1i2} \text{LF}_{3,1,1,-2} [\mathbf{m}_2, \mathbf{m}_q^{i1}, \tilde{\mu}] + \frac{5}{16} \mathbf{g}_1 \mathbf{g}_2^2 y_u^{i1i2} \text{LF}_{2,2,1,-2} [\mathbf{m}_2, \tilde{\mu}, \mathbf{m}_q^{i1}] - \\
& \frac{5}{9} \mathbf{g}_1 \mathbf{m}_3 \mathbf{g}_3^2 a_u^{i1i2} \text{LF}_{2,1,1,0} [\mathbf{m}_3, \mathbf{m}_q^{i1}, \mathbf{m}_{\bar{u}}^{i2}] + \frac{1}{6} \mathbf{g}_1 \mathbf{m}_3 \mathbf{g}_3^2 a_u^{i1i2} \text{LF}_{2,2,1,-1} [\mathbf{m}_3, \mathbf{m}_q^{i1}, \mathbf{m}_{\bar{u}}^{i2}] + \\
& \frac{5}{9} \mathbf{g}_1 \mathbf{m}_3 \mathbf{g}_3^2 a_u^{i1i2} \text{LF}_{3,1,1,-1} [\mathbf{m}_3, \mathbf{m}_q^{i1}, \mathbf{m}_{\bar{u}}^{i2}] - \frac{1}{6} \mathbf{g}_1 \mathbf{m}_3 \mathbf{g}_3^2 a_u^{i1i2} \text{LF}_{2,2,1,-1} [\mathbf{m}_3, \mathbf{m}_{\bar{u}}^{i2}, \mathbf{m}_q^{i1}] + \\
& \frac{1}{16} \mathbf{g}_1 \tilde{\mu}^2 \overline{y_d^{\text{pr}}} y_d^{i1r} y_u^{pi2} \text{LF}_{2,2,1,-1} [\mathbf{m}_d^r, \tilde{\mu}, \mathbf{m}_q^p] - \\
& \frac{1}{16} \mathbf{g}_1 \tilde{\mu}^2 \overline{y_d^{\text{pr}}} y_d^{i1r} y_u^{pi2} \text{LF}_{2,2,1,-1} [\mathbf{m}_q^p, \tilde{\mu}, \mathbf{m}_d^r] + \frac{5}{72} \mathbf{g}_1^3 y_u^{i1i2} \text{LF}_{2,1,1,-1} [\tilde{\mu}, \mathbf{m}_1, \mathbf{m}_q^{i1}] - \\
& \frac{1}{18} \mathbf{g}_1^3 y_u^{i1i2} \text{LF}_{3,1,1,-2} [\tilde{\mu}, \mathbf{m}_1, \mathbf{m}_q^{i1}] - \frac{5}{18} \mathbf{g}_1^3 y_u^{i1i2} \text{LF}_{2,1,1,-1} [\tilde{\mu}, \mathbf{m}_1, \mathbf{m}_{\bar{u}}^{i2}] + \\
& \frac{1}{18} \mathbf{g}_1^3 y_u^{i1i2} \text{LF}_{3,1,1,-2} [\tilde{\mu}, \mathbf{m}_1, \mathbf{m}_{\bar{u}}^{i2}] - \frac{5}{8} \mathbf{g}_1 \mathbf{g}_2^2 y_u^{i1i2} \text{LF}_{2,1,1,-1} [\tilde{\mu}, \mathbf{m}_2, \mathbf{m}_q^{i1}] + \\
& \frac{1}{2} \mathbf{g}_1 \mathbf{g}_2^2 y_u^{i1i2} \text{LF}_{3,1,1,-2} [\tilde{\mu}, \mathbf{m}_2, \mathbf{m}_q^{i1}] + \frac{1}{24} \mathbf{g}_1 \tilde{\mu}^2 \overline{y_d^{\text{pr}}} y_d^{i1r} y_u^{pi2} \text{LF}_{2,1,1,0} [\tilde{\mu}, \mathbf{m}_d^r, \mathbf{m}_q^p] + \\
& \frac{5}{24} \mathbf{g}_1 \tilde{\mu}^2 \overline{y_d^{\text{pr}}} y_d^{i1r} y_u^{pi2} \text{LF}_{3,1,1,-1} [\tilde{\mu}, \mathbf{m}_d^r, \mathbf{m}_q^p] \Big)
\end{aligned}$$